INDIVIDUAL WEEKLY PROJECT REPORTS

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Role: Data Collection and Data Quality
Project Title: Flight Delay Analysis and Visualization

WEEK 3 – Data Quality Assessment

During the third week, I focused on assessing the quality of the collected flight dataset. Before the data could be used for detailed analysis, it was important to examine whether the records were complete, consistent, and suitable for processing.

I explored the dataset for possible missing values and examined which columns contained incomplete information. Missing values can affect statistical calculations and may create difficulties during visualization, so identifying them early was an important task.

I also checked for duplicate records that could potentially affect the accuracy of the analysis. Duplicate entries may cause certain observations to be counted more than once, resulting in misleading summaries.

Another important part of my work involved examining data types and identifying inconsistencies in the representation of variables. I understood that numerical fields should be stored in suitable numerical formats, while categorical fields should be represented appropriately.

I shared my observations with Annalakshmi, who was responsible for preprocessing, so that the identified issues could be examined and addressed during the cleaning stage.

Outcome and Learning:

This week helped me understand the importance of data quality assessment. I learned that checking the completeness, consistency, and accuracy of a dataset is essential before performing any meaningful analysis.